COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 212	**SECTION:**	CPE31S3
DATE PERFORMED:	10/06/2026	**DATE SUBMITTED:**	10/06/2026
NAME:	Pios, Joshua Paul B.	**INSTRUCTOR:**	Engr. Jimlord Quejado

Seatwork 10.1 **Install, Configure and Manage Enterprise Log Monitoring via Ansible**

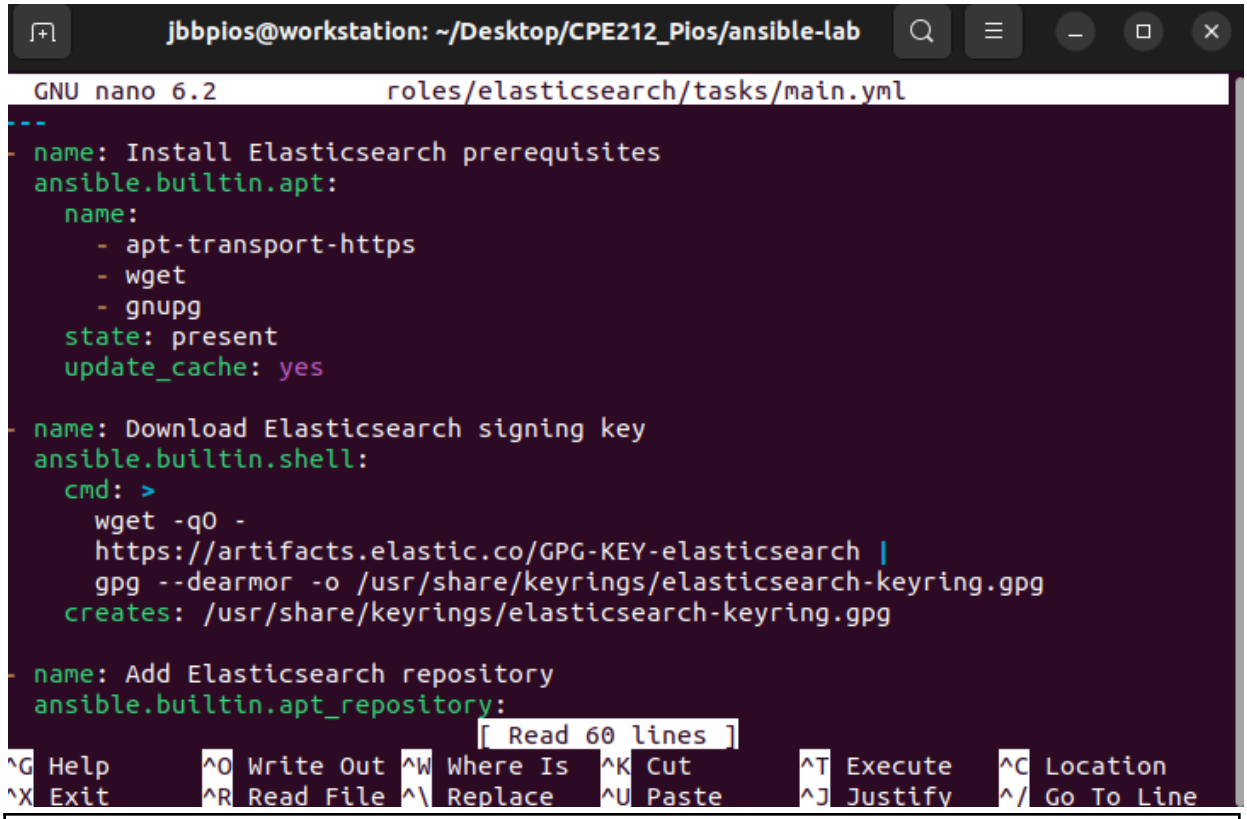
1. PROCEDURE OUTPUT

PART 1: Creating Elasticsearch Role

This part creates the Ansible role responsible for installing and configuring Elasticsearch. Elasticsearch serves as the main storage and search component for the log-monitoring system.

```
mkdir -p roles/elasticsearch/tasks
mkdir -p roles/elasticsearch/handlers
mkdir -p roles/elasticsearch/templates
```

Create Elasticsearch tasks




```
---
- name: Install Elasticsearch prerequisites
```

```
  ansible.builtin.apt:
    name:
      - apt-transport-https
      - wget
      - gnupg
    state: present
    update_cache: true

- name: Download Elasticsearch signing key
  ansible.builtin.shell:
    cmd: >
      wget -qO -
      https://artifacts.elastic.co/GPG-KEY-elasticsearch |
      gpg --dearmor -o /usr/share/keyrings/elasticsearch-keyring.gpg
    creates: /usr/share/keyrings/elasticsearch-keyring.gpg

- name: Add Elasticsearch repository
  ansible.builtin.apt_repository:
    repo: "deb [signed-by=/usr/share/keyrings/elasticsearch-keyring.gpg]
https://artifacts.elastic.co/packages/9.x/apt stable main"
    state: present
    filename: elastic-9.x

- name: Install Elasticsearch
  ansible.builtin.apt:
    name: elasticsearch
    state: present
    update_cache: true

- name: Create Elasticsearch data directory
  ansible.builtin.file:
    path: /usr/share/elasticsearch/data
    state: directory
    owner: elasticsearch
    group: elasticsearch
    mode: '0750'

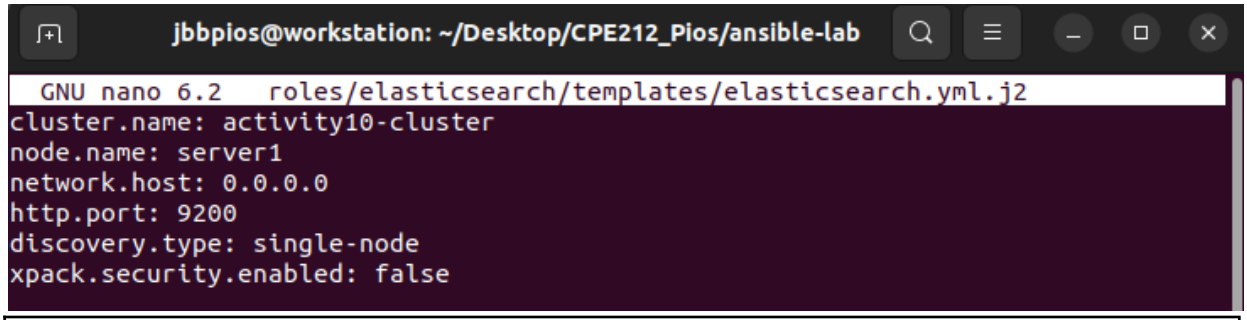
- name: Create Elasticsearch logs directory
  ansible.builtin.file:
    path: /usr/share/elasticsearch/logs
    state: directory
    owner: elasticsearch
    group: elasticsearch
    mode: '0750'

- name: Configure Elasticsearch
  ansible.builtin.template:
    src: elasticsearch.yml.j2
```

```
    dest: /etc/elasticsearch/elasticsearch.yml
    owner: root
    group: elasticsearch
    mode: '0660'
  notify: Restart Elasticsearch

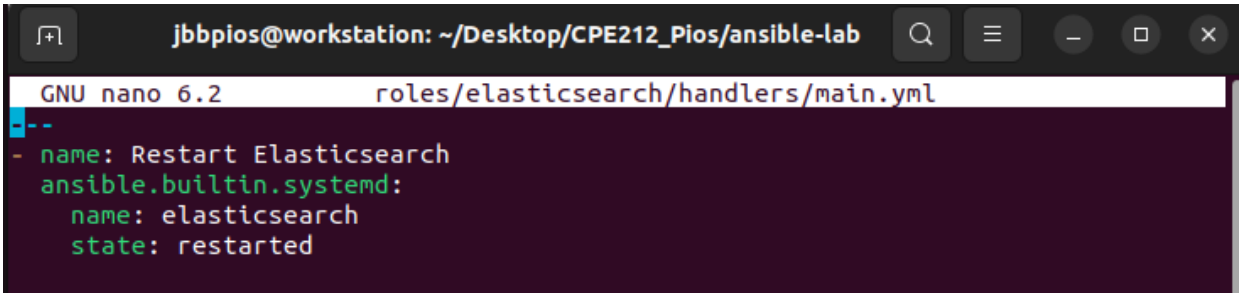
- name: Enable and start Elasticsearch
  ansible.builtin.systemd:
    name: elasticsearch
    enabled: true
    state: started
```

Create the template




```
cluster.name: activity10-cluster
node.name: server1
network.host: 0.0.0.0
http.port: 9200
discovery.type: single-node
xpack.security.enabled: false
```

Create the handler




```
---
- name: Restart Elasticsearch
  ansible.builtin.systemd:
    name: elasticsearch
    state: restarted
```

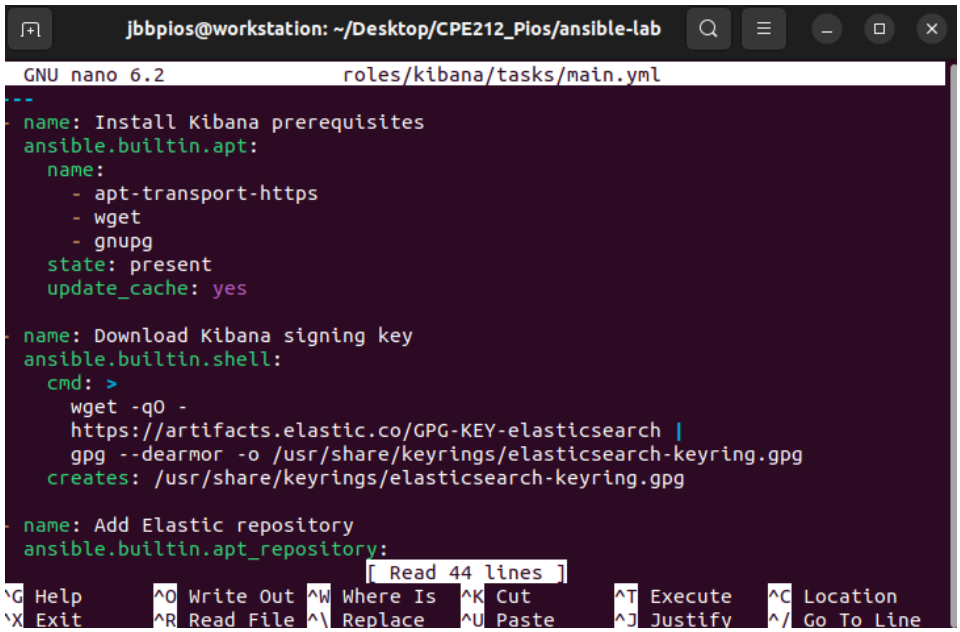
PART 2: Creating Kibana Role

This part creates the Ansible role for Kibana, which provides the web-based interface used to interact with and visualize information stored in Elasticsearch.

Create directories

```
mkdir -p roles/kibana/tasks
mkdir -p roles/kibana/handlers
mkdir -p roles/kibana/templates
```

Create Kibana tasks




```
---
- name: Install Kibana prerequisites
  ansible.builtin.apt:
    name:
      - apt-transport-https
      - wget
      - gnupg
    state: present
    update_cache: true

- name: Download Kibana signing key
  ansible.builtin.shell:
    cmd: >
      wget -qO -
      https://artifacts.elastic.co/GPG-KEY-elasticsearch |
      gpg --dearmor -o /usr/share/keyrings/elasticsearch-keyring.gpg
```

```
    creates: /usr/share/keyrings/elasticsearch-keyring.gpg

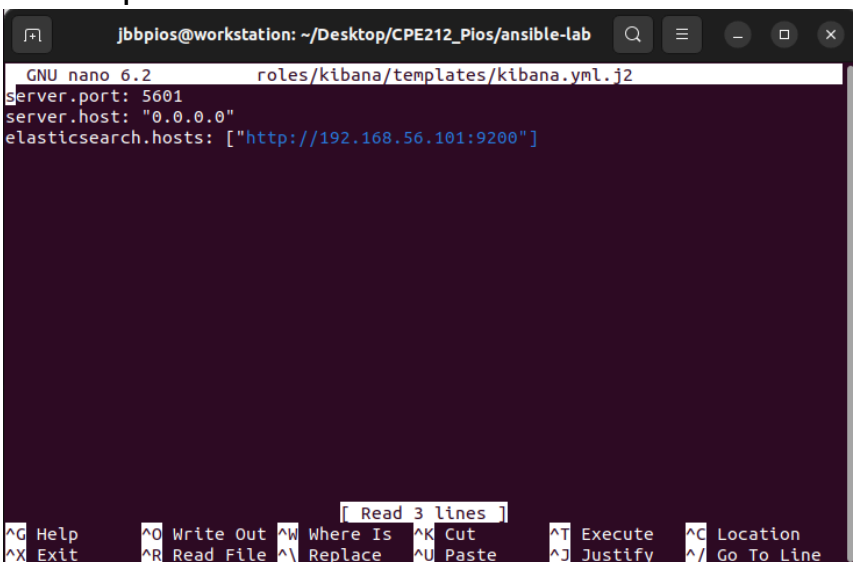
- name: Add Elastic repository
  ansible.builtin.apt_repository:
    repo: "deb [signed-by=/usr/share/keyrings/elasticsearch-keyring.gpg]
https://artifacts.elastic.co/packages/9.x/apt stable main"
    state: present
    filename: elastic-9.x

- name: Install Kibana
  ansible.builtin.apt:
    name: kibana
    state: present
    update_cache: true

- name: Configure Kibana
  ansible.builtin.template:
    src: kibana.yml.j2
    dest: /etc/kibana/kibana.yml
    owner: root
    group: kibana
    mode: '0660'
  notify: Restart Kibana

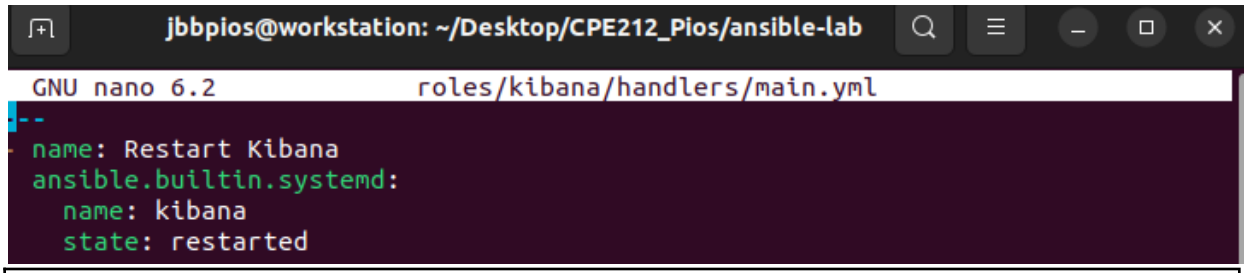
- name: Enable and start Kibana
  ansible.builtin.systemd:
    name: kibana
    enabled: true
    state: started
```

Create template



```
server.port: 5601
server.host: "0.0.0.0"
elasticsearch.hosts: ["http://192.168.56.101:9200"]
```

Create handler




```
---
- name: Restart Kibana
  ansible.builtin.systemd:
    name: kibana
    state: restarted
```

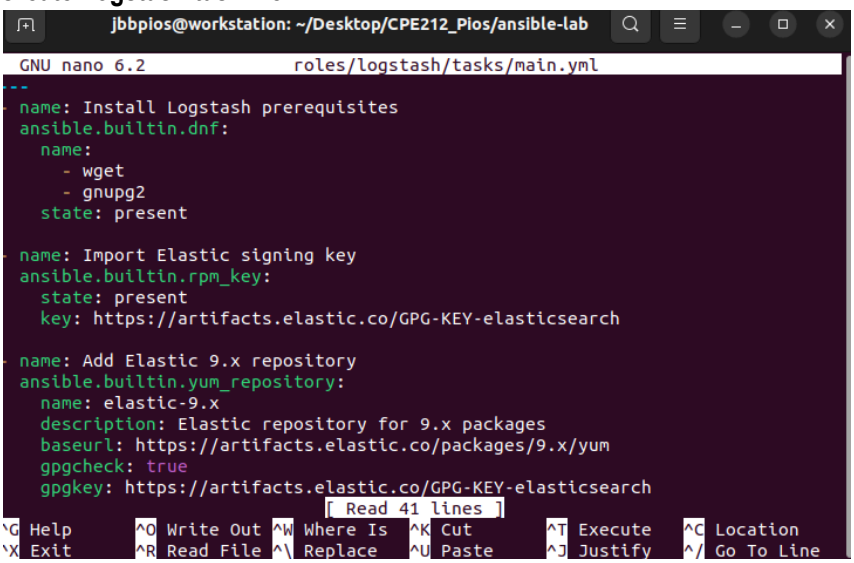
PART 3: Create Logstash Role

This part creates the Logstash role. Logstash receives log data, processes it, and forwards it to Elasticsearch for storage and analysis.

Create directories

```
mkdir -p roles/logstash/tasks
mkdir -p roles/logstash/handlers
mkdir -p roles/logstash/templates
```

Create Logstash task file



```
---
- name: Install Logstash prerequisites
  ansible.builtin.dnf:
    name:
      - wget
      - gnupg2
    state: present

- name: Import Elastic signing key
  ansible.builtin.rpm_key:
    state: present
    key: https://artifacts.elastic.co/GPG-KEY-elasticsearch

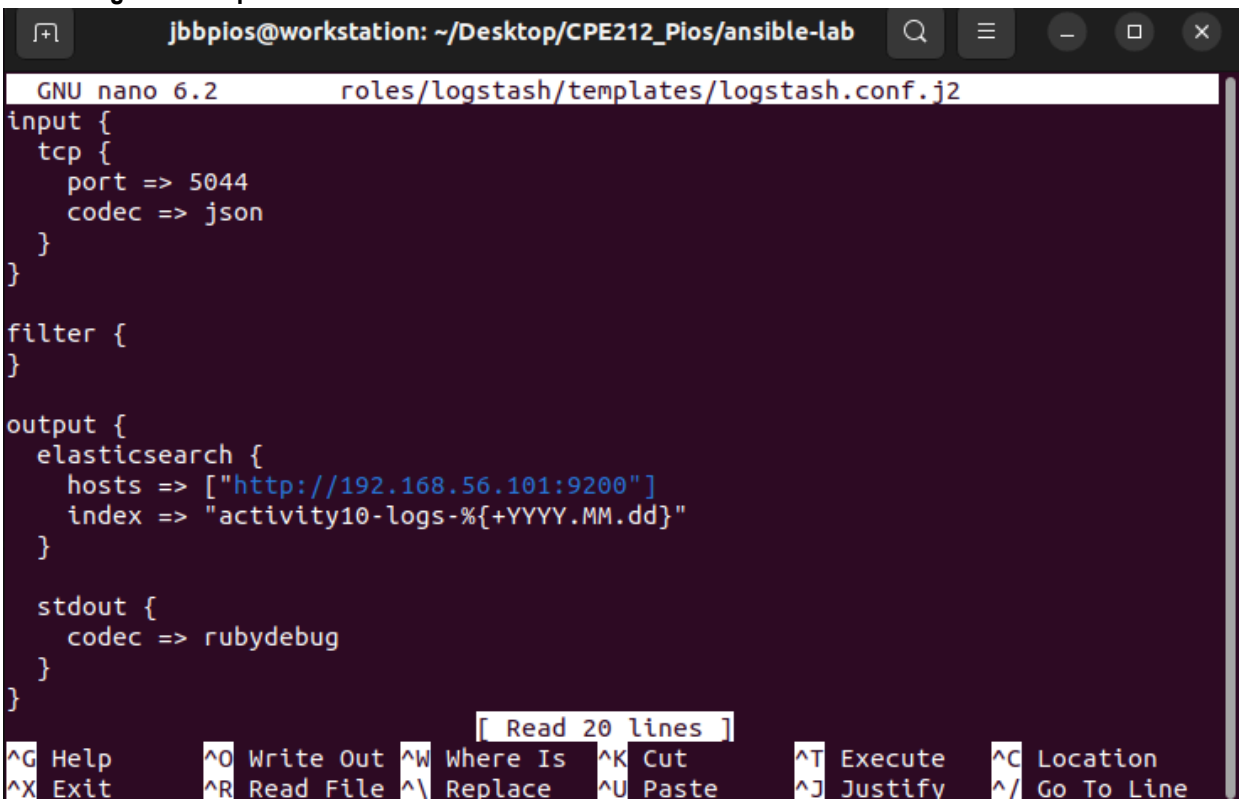
- name: Add Elastic 9.x repository
  ansible.builtin.yum_repository:
    name: elastic-9.x
    description: Elastic repository for 9.x packages
    baseurl: https://artifacts.elastic.co/packages/9.x/yum
    gpgcheck: true
    gpgkey: https://artifacts.elastic.co/GPG-KEY-elasticsearch
    enabled: true

- name: Install Logstash
  ansible.builtin.dnf:
    name: logstash
    state: present

- name: Configure Logstash
  ansible.builtin.template:
    src: logstash.conf.j2
    dest: /etc/logstash/conf.d/activity10.conf
    owner: root
    group: root
    mode: '0644'
  notify: Restart Logstash

- name: Enable and start Logstash
  ansible.builtin.systemd:
    name: logstash
    enabled: true
    state: started
```

Create Logstash template




```
input {
  tcp {
    port => 5044
    codec => json
  }
}

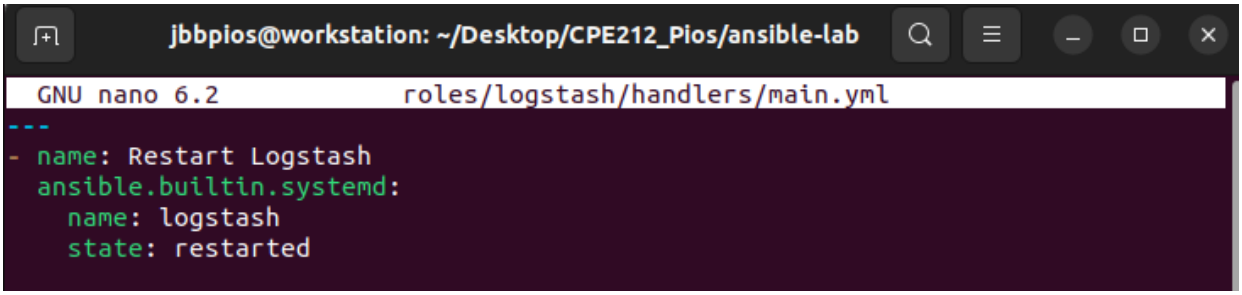
filter {
}

output {
  elasticsearch {
    hosts => ["http://192.168.56.101:9200"]
    index => "activity10-logs-%{+YYYY.MM.dd}"
  }

  stdout {
```

```
    codec => rubydebug
  }
}
```

Create handler

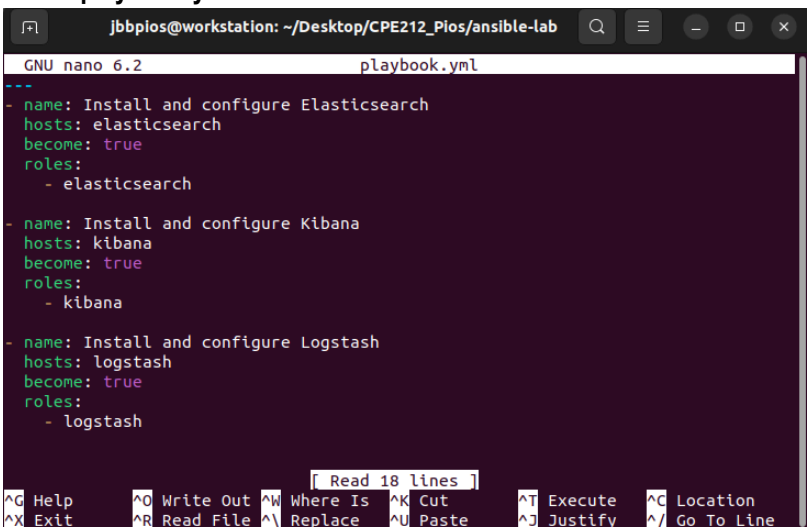



```
---
- name: Restart Logstash
  ansible.builtin.systemd:
    name: logstash
    state: restarted
```

PART 4: Create the Main Playbook

The main playbook connects all three roles and determines which role should be executed on each designated host.

Create playbook.yml




```
---
- name: Install and configure Elasticsearch
  hosts: elasticsearch
  become: true
```

```
  roles:
    - elasticsearch

- name: Install and configure Kibana
  hosts: kibana
  become: true
  roles:
    - kibana

- name: Install and configure Logstash
  hosts: logstash
  become: true
  roles:
    - logstash
```

Run everything

```
(base) jbbpios@workstation:~/Desktop/CPE212_Pios/ansible-lab$ ansible-playbook p
laybook.yml -K
BECOME password:

PLAY [Prepare all Elastic Stack servers] ***********************************

TASK [Gathering Facts] *****************************************************
ok: [server2]
ok: [server1]
ok: [server3]

TASK [common : Install common prerequisites on Ubuntu] *********************
skipping: [server3]
ok: [server2]
ok: [server1]

TASK [common : Install common prerequisites on CentOS] *********************
skipping: [server1]
skipping: [server2]
ok: [server3]

TASK [common : Download Elastic signing key on Ubuntu] *********************
skipping: [server3]

PLAY RECAP *****************************************************************
server1                    : ok=18   changed=0    unreachable=0    failed=0    s
kipped=3    rescued=0    ignored=0
server2                    : ok=18   changed=0    unreachable=0    failed=0    s
kipped=3    rescued=0    ignored=0
server3                    : ok=19   changed=2    unreachable=0    failed=0    s
kipped=3    rescued=0    ignored=0
```

```
ansible-playbook playbook.yml -K
```

PART 5: Verification

```
(base) jbbpios@workstation:~/Desktop/CPE212_Pios/ansible-lab$ ansible all -m she
ll -a "systemctl is-active elasticsearch" --become -K
BECOME password:
server2 | CHANGED | rc=0 >>
active
server1 | CHANGED | rc=0 >>
active
```

```
(base) jbbpios@workstation:~/Desktop/CPE212_Pios/ansible-lab$ ansible all -m she
ll -a "systemctl is-active kibana" --become -K
BECOME password:
server2 | CHANGED | rc=0 >>
active
server1 | CHANGED | rc=0 >>
active
server3 | CHANGED | rc=0 >>
active
(base) jbbpios@workstation:~/Desktop/CPE212_Pios/ansible-lab$
```

```
(base) jbbpios@workstation:~/Desktop/CPE212_Pios/ansible-lab$ ansible all -m she
ll -a "systemctl is-active logstash" --become -K
BECOME password:
server1 | CHANGED | rc=0 >>
active
server2 | CHANGED | rc=0 >>
```

```
(base) jbbpios@workstation:~/Desktop/CPE212_Pios/ansible-lab$ ansible all -m she
ll -a "ss -lntp | grep ':9200'" --become -K
BECOME password:
server1 | CHANGED | rc=0 >>
LISTEN 0      4096                    *:9200            *:*    users:(("java",pi
d=5183,fd=881))


server2 | CHANGED | rc=0 >>
LISTEN 0      4096                    *:9200            *:*    users:(("java",pi
d=11696,fd=881))
server3 | CHANGED | rc=0 >>
LISTEN 0      4096                *:9200            *:*    users:(("java",pid=306
1,fd=878))
```

```
(base) jbbpios@workstation:~/Desktop/CPE212_Pios/ansible-lab$ ansible all -m she
ll -a "ss -lntp | grep ':5601'" --become -K
BECOME password:
server1 | CHANGED | rc=0 >>
LISTEN 0      511                0.0.0.0:5601      0.0.0.0:*    users:(("MainThre
ad",pid=13253,fd=28))


server2 | CHANGED | rc=0 >>
LISTEN 0      511                0.0.0.0:5601      0.0.0.0:*    users:(("MainThre
ad",pid=13011,fd=159))
server3 | CHANGED | rc=0 >>
LISTEN 0      511           0.0.0.0:5601      0.0.0.0:*    users:(("MainThread",p
id=1237,fd=18))
```

1. What are the benefits of having a log monitoring tool?

Having a log monitoring tool provides a centralized way to collect, store, and monitor system logs. It helps administrators detect errors, unusual activities, and possible security problems more easily. In this activity, Elasticsearch, Logstash, and Kibana work together to collect and manage logs, making it easier to analyze system activity and troubleshoot problems. Overall, log monitoring improves system visibility, security, and reliability.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Conclusion

This activity demonstrated how Ansible can be used to install, configure, and manage an enterprise log monitoring system using Elasticsearch, Kibana, and Logstash. By creating separate Ansible roles and combining them in a main playbook, the installation and configuration process becomes more organized and automated. The activity also showed how centralized log monitoring can help administrators observe system activities, identify problems, and manage logs more efficiently.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: N/A

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: N/A

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: N/A

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

Lab activities, templates, and related instructional materials shall not be reuploaded, reproduced, distributed, or shared with external organizations or third parties without the prior express written permission of the Technological Institute of the Philippines.